

Experiment 23

Title

Write a Prolog Program to implement a Family Tree.

Aim

To implement a Family Tree in Prolog using facts and rules to identify different family relationships.

Objective

To understand how Prolog uses facts and rules to represent and retrieve family relationships such as **mother, father, grandfather, grandmother, sister, and brother**.

Algorithm

Step 1: Define facts for male and female family members.

Step 2: Define parent relationships between the family members.

Step 3: Create rules for **mother** and **father** using gender and parent facts.

Step 4: Create rules for **grandfather, grandmother, sister, and brother**.

Step 5: Execute queries to retrieve the required family relationships.

Program

% Family Members

female(pam).

female(liz).

female(ann).

female(pat).

male(tom).

male(bob).

male(jim).

% Parent Relations

parent(tom, liz).

```
parent(tom, bob).  
parent(pam, liz).  
parent(pam, bob).  
parent(bob, ann).  
parent(bob, pat).  
parent(liz, jim).
```

```
% Mother
```

```
mother(X, Y) :-  
    female(X),  
    parent(X, Y).
```

```
% Father
```

```
father(X, Y) :-  
    male(X),  
    parent(X, Y).
```

```
% Grandmother
```

```
grandmother(X, Y) :-  
    female(X),  
    parent(X, Z),  
    parent(Z, Y).
```

```
% Grandfather
```

```
grandfather(X, Y) :-
```

```
male(X),  
parent(X, Z),  
parent(Z, Y).
```

% Sister

```
sister(X, Y) :-  
    female(X),  
    parent(Z, X),  
    parent(Z, Y),  
    X \= Y.
```

% Brother

```
brother(X, Y) :-  
    male(X),  
    parent(Z, X),  
    parent(Z, Y),  
    X \= Y.
```

Sample Input

?- father(X, bob).

?- mother(X, bob).

?- grandfather(X, ann).

?- grandmother(X, ann).

?- sister(X, pat).

?- brother(X, liz).

Sample Output

```
consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp23.pl').
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp23.pl for byte code...
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp23.pl compiled, 61 lines read - 3759 bytes written, 5 ms

(16 ms) yes
| ?- father(X, bob).

X = tom ?

(109 ms) yes
| ?- mother(X, bob).

X = pam ?

(63 ms) yes
| ?- grandfather(X, ann).

X = tom ?

yes
| ?- grandmother(X, ann).

X = pam ?

(47 ms) yes
| ?- sister(X, pat).

X = ann ?

yes
| ?- brother(X, liz).

X = bob ?

(47 ms) yes
| ?-
```

Result

Hence, the Prolog program to implement a **Family Tree** was executed successfully and the correct family relationships were obtained.